VIJAYVENKATESH JAYARAJ

AI ENGINEER
Plainsboro, NJ | +1 7745256102 | vijayvenkateshj012@gmail.com | LinkedIn | GitHub

Summary

AI Engineer with 4.5 years in software and data engineering, integrating pretrained models into AI agents and document pipelines deployed for clients, made reliable by evaluation, guardrails and human review. Started in conversational AI, designing and deploying 50+ Rasa chatbots during a 15-month internship.

PROFESSIONAL EXPERIENCE

Princeton IT Services | *AI Engineer* **May 2025 - Present**

- Built a Copilot Studio label-review agent for a food brand that runs in production, pulls label files from SharePoint through Microsoft Graph, detects ingredient, Nutrition Facts, and badge changes using OCR and OpenCV on Azure Functions with Python, and returns annotated PDFs, reducing label review time
- Raised change-detection accuracy from 68% to 88% on a self-built 50-case ground-truth benchmark, using RapidOCR on ONNX Runtime and OpenCV, with confidence thresholds that flag unclear reads instead of guessing.
- Designed an on-device insurance intake agent that reads five-document broker packets, including ACORD forms, loss runs, and email, using PyMuPDF, pdfplumber, RapidOCR, and local LLMs; routed uncertain values to external models for human-in-the-loop review, improving intake reliability while limiting unsupported outputs
- Ran Qwen3-30B on an ASUS Ascent GX10 at ~30 tokens/s and built a Foundry Local-compatible bridge to Ollama/llama.cpp (CUDA), cutting six model calls from ~10 min on CPU to 62.6 s.
- Added enterprise governance to a Claude-powered regulatory agent: prompts and responses are audited in Microsoft Purview DSPM for AI via Microsoft Graph, with per-turn telemetry in App Insights queried using KQL.
- Built Microsoft 365 and Copilot risk monitoring for a client: 9 data-sync jobs load file-sharing, access and AI-agent activity into PostgreSQL, helping remove 1,250 risky sharing links and save 40 hrs/month of access reviews.
- Automated weekly AI code reviews across 25+ repositories with Claude Code and GitHub Actions, flagging 50+ bugs and security risks and saving more than 5 hours per week of manual review.

Tata Consultancy Services | *Assistant System Engineer - Data & ML* **May 2022 - Nov 2023**

- Developed a Demand Forecasting and Inventory Optimization project for a retail company in 2022 using Python, SQL, Pandas, and Scikit-learn to predict product demand and support supply chain planning.
- Built Azure Data Factory ETL pipelines to integrate and clean sales and inventory data, enabling forecasting of seasonality, promotions, and purchasing trends.
- Trained and evaluated XGBoost and Random Forest forecasting models using time series cross-validation, MAE, and RMSE; created Power BI dashboards to present demand forecasts and identify potential stockouts and excess inventory.
- Engineered time-based and lag features (rolling averages, holiday flags, promotion indicators) and applied hyperparameter tuning with GridSearchCV, improving forecast accuracy and reducing MAE by an estimated 12–15% over baseline models
- Calculated safety stock levels and reorder points from forecasted demand and lead-time variability, helping the business reduce excess inventory and lower stockout risk across key product categories.

Kenyt.ai | *Conversation Designer Intern* **Jan 2020 - Mar 2021**

- Designed, built and deployed 50+ conversational AI chatbots across real estate, healthcare, education, restaurants and e-commerce, modeling intents, entities, training utterances and dialogue flows in Rasa.
- Built reusable templates from JSON dialogue flows and knowledge bases, cutting client onboarding from 2 weeks to 3 days; created workflow modules (FAQs, payments, booking) reused across 100+ bots via REST APIs and webhooks.
- Ran QA, regression testing and UAT with scripted conversations and Postman; weekly log reviews raised the answer rate by 20%.

TECHNICAL SKILLS

- **AI & LLM Integration**: Claude, Claude Code, Copilot Studio, Qwen3, Ollama, llama.cpp (CUDA), Foundry Local, Rasa, Dialogflow
- **Document AI & Vision**: OCR (RapidOCR), ONNX Runtime, OpenCV, PyMuPDF, pdfplumber
- **Evaluation & Governance**: ground-truth benchmarking, confidence thresholds, human-in-the-loop review, regression testing, UAT, Azure Application Insights, KQL, Microsoft Purview
- **ML Model Training**: scikit-learn, logistic regression, random forest, XGBoost, feature engineering, cross-validation, ROC-AUC
- **Engineering & Cloud**: Python, pandas, TypeScript, JavaScript, FastAPI, REST APIs, webhooks, Postman, Azure Functions, Azure Data Factory, Microsoft Graph API, PostgreSQL, SQL, Spark (Scala), Databricks, Power BI, GitHub Actions

PROJECTS

Diabetes Patient Readmission Prediction | *Clark University* **Jan 2024 - May 2024**

- Trained Logistic Regression and Random Forest classifiers in scikit-learn (Random Forest: 0.89 ROC-AUC, 84% accuracy).
- Collaborated on feature engineering to identify readmission risk factors; evaluated models with cross-validation.

EDUCATION

Clark University **2024 - 2025**

Master of Science, Business Analytics (GPA: 3.7)

- **Coursework:** Machine Learning, Advanced Big Data Computing & Programming, Analytics Programming, Intermediate Statistical Modeling for Analytics

Anna University **2018 - 2022**

Bachelor of Engineering, Computer Science (GPA: 3.8)

- **Coursework:** Design and Analysis of Algorithms, Probability, Statistics and Queuing Theory